Exercise 2.3.2. Define a map $R : \mathbb{T}\times\mathbb{T} \to \mathbb{T}\times\mathbb{T}$ by $R(x,y) = (x+\alpha, y+\alpha)$ for an irrational α. Show that for any set of the form $A \times B$ with A, B measurable subsets of $\mathbb{T}$ (such a set is called a *measurable rectangle*) has the property of Definition 2.13, but the transformation R is not ergodic, even if α is irrational.

Exercise 2.3.3. (a) Find an arithmetic condition on α_1 and α_2 that is equivalent to the ergodicity of $R_{\alpha_1}\times R_{\alpha_2} : \mathbb{T}\times\mathbb{T} \to \mathbb{T}\times\mathbb{T}$ with respect to $m_{\mathbb{T}}\times m_{\mathbb{T}}$.
(b) Generalize part (a) to characterize ergodicity of the rotation

$$R_{\alpha_1}\times\cdots\times R_{\alpha_n} : \mathbb{T}^n \to \mathbb{T}^n$$

with respect to $m_{\mathbb{T}^n}$.

Exercise 2.3.4. Prove that any factor of an ergodic measure-preserving system is ergodic.

Exercise 2.3.5. Extend Proposition 2.14 by showing that for each $p \in [1,\infty]$ a measure-preserving transformation T is ergodic if and only if for any L^p function f, $f\circ T = f$ almost everywhere implies that f is almost everywhere equal to a constant.

Exercise 2.3.6. Strengthen Proposition 2.14(5) by showing that a measure-preserving transformation T is ergodic if and only if any measurable function $f : X \to \mathbb{R}$ with $f(Tx) \geqslant f(x)$ almost everywhere is equal to a constant almost everywhere.

Exercise 2.3.7. Let X be a compact metric space and let $T : X \to X$ be continuous. Suppose that μ is a T-invariant ergodic probability measure defined on the Borel subsets of X. Prove that for μ-almost every $x \in X$ and every y in the support of μ there exists a sequence $n_k \nearrow \infty$ such that $T^{n_k}(x) \to y$ as $k \to \infty$. Here the support $\operatorname{Supp}(\mu)$ of μ is the smallest closed subset A of X with $\mu(A) = 1$; alternatively

$$\operatorname{Supp}(\mu) = X \smallsetminus \bigcup_{\substack{O\subseteq X \text{ open},\\ \mu(O)=0}} O.$$

Notice that X has a countable base for its topology, so the union is still a μ-null set (see p. 406).

2.4 Associated Unitary Operators

A different kind of action[16] induced by a measure-preserving map T on a function space is the *associated operator* $U_T : L^2_\mu \to L^2_\mu$ defined by

$$U_T(f) = f\circ T.$$

Recall that L^2_μ is a Hilbert space, and for any functions $f_1, f_2 \in L^2_\mu$,

$$\begin{aligned}\langle U_T f_1, U_T f_2\rangle &= \int f_1 \circ T \cdot \overline{f_2 \circ T}\,\mathrm{d}\mu \\ &= \int f_1 \overline{f_2}\,\mathrm{d}\mu \qquad \text{(since } \mu \text{ is } T\text{-invariant)} \\ &= \langle f_1, f_2\rangle\,.\end{aligned}$$

Here it is natural to think of functions as being complex-valued; it will be clear from the context when members of L^2_μ are allowed to be complex-valued. Thus U_T is an isometry mapping L^2_μ into L^2_μ whenever $(X, \mathscr{B}_X, \mu, T)$ is a measure-preserving system.

If $U : \mathscr{H}_1 \to \mathscr{H}_2$ is a continuous linear operator from one Hilbert space to another then the relation

$$\langle Uf, g\rangle = \langle f, U^* g\rangle$$

defines an associated operator $U^* : \mathscr{H}_2 \to \mathscr{H}_1$ called the *adjoint* of U. The operator U is an *isometry* (that is, has $\|Uh\|_{\mathscr{H}_2} = \|h\|_{\mathscr{H}_1}$ for all $h \in \mathscr{H}_1$) if and only if

$$U^* U = I_{\mathscr{H}_1} \tag{2.5}$$

is the identity operator on $\mathscr{H}_1$ and

$$UU^* = P_{\operatorname{Im} U} \tag{2.6}$$

is the projection operator onto $\operatorname{Im} U$. Finally, an invertible linear operator U is called *unitary* if $U^{-1} = U^*$, or equivalently if U is invertible and

$$\langle Uh_1, Uh_2\rangle = \langle h_1, h_2\rangle \tag{2.7}$$

for all $h_1, h_2 \in \mathscr{H}_1$. If $U : \mathscr{H}_1 \to \mathscr{H}_2$ satisfies (2.7) then U is an isometry (even if it is not invertible). Thus for any measure-preserving transformation T, the associated operator U_T is an isometry, and if T is invertible then the associated operator U_T is a unitary operator, called the *associated unitary operator* of T or *Koopman operator* of T.

A property of a measure-preserving transformation is said to be a *spectral* or *unitary property* if it can be detected by studying the associated operator on L^2_μ.

Lemma 2.18. *A measure-preserving transformation T is ergodic if and only if* 1 *is a simple eigenvalue of the associated operator U_T. Hence ergodicity is a unitary property.*

Proof. This follows from the proof of the equivalence of (2) and (5) in Proposition 2.14 or via Exercise 2.3.5 applied with $p = 2$: an eigenfunction for the eigenvalue 1 is a T-invariant function, and ergodicity is characterized by the property that the only T-invariant functions are the constants. □

An isometry $U : \mathscr{H}_1 \to \mathscr{H}_2$ between Hilbert spaces[(17)] sends the expansion of an element

$$x = \sum_{n=1}^{\infty} c_n e_n$$

in terms of a complete orthonormal basis $\{e_n\}$ for $\mathscr{H}_1$ to a convergent expansion

$$U(x) = \sum_{n=1}^{\infty} c_n U(e_n)$$

in terms of the orthonormal set $\{U(e_n)\}$ in $\mathscr{H}_2$.

We will use this observation to study ergodicity of some of the examples using harmonic analysis rather than the geometrical arguments used earlier in this chapter.

Proof of Proposition 2.16 by Fourier analysis. Assume that α is irrational and let $f \in L^2(\mathbb{T})$ be a function invariant under R_α. Then f has a Fourier expansion $f(t) = \sum_{n\in\mathbb{Z}} c_n \mathrm{e}^{2\pi i n t}$ (both equality and convergence are meant in $L^2(\mathbb{T})$). Now f is invariant, so $\|f \circ R_\alpha - f\|_2 = 0$. By uniqueness of Fourier coefficients, this requires that $c_n = c_n \mathrm{e}^{2\pi i n\alpha}$ for all $n \in \mathbb{Z}$. Since α is irrational, $\mathrm{e}^{2\pi i n \alpha}$ is only equal to 1 when $n = 0$, so this equation forces c_n to be 0 except when $n = 0$. Thus f is a constant almost everywhere, and hence R_α is ergodic.

If $\alpha \in \mathbb{Q}$ then write $\alpha = \frac{p}{q}$ in lowest terms. The function $g(t) = \mathrm{e}^{2\pi \mathrm{i} q t}$ is invariant under R_α but is not equal almost everywhere to a constant. □

Similar methods characterize ergodicity for endomorphisms.

Proof of Proposition 2.17 by Fourier analysis. Let $f \in L^2(\mathbb{T})$ be a function with $f \circ T_2 = f$ (equalities again are meant as elements of $L^2(\mathbb{T})$). Then f has a Fourier expansion $f(t) = \sum_{n\in\mathbb{Z}} c_n \mathrm{e}^{2\pi \mathrm{i} n t}$ with

$$\sum_{n\in\mathbb{Z}} |c_n|^2 = \|f\|_2^2 < \infty. \tag{2.8}$$

By invariance under T_2,

$$f(T_2 t) = \sum_{n\in\mathbb{Z}} c_n \mathrm{e}^{2\pi \mathrm{i} 2 n t} = f(t) = \sum_{n\in\mathbb{Z}} c_n \mathrm{e}^{2\pi \mathrm{i} n t},$$

so by uniqueness of Fourier coefficients we must have $c_{2n} = c_n$ for all $n \in \mathbb{Z}$. If there is some $n \neq 0$ with $c_n \neq 0$ then this contradicts (2.8), so we deduce that $c_n = 0$ for all $n \neq 0$. It follows that f is constant a.e., so T_2 is ergodic. □

The same argument gives the general abelian case, where Fourier analysis is replaced by character theory (see Sect. C.3 for the background). Notice that for a character $\chi : X \to \mathbb{S}^1$ on a compact abelian group and a continuous homomorphism $T : X \to X$, the map $\chi \circ T : X \to \mathbb{S}^1$ is also a character on X.

Theorem 2.19. *Let $T : X \to X$ be a continuous surjective homomorphism of a compact abelian group X. Then T is ergodic with respect to the Haar measure m_X if and only if the identity $\chi(T^n x) = \chi(x)$ for some $n > 0$ and character $\chi \in \widehat{X}$ implies that χ is the trivial character with $\chi(x) = 1$ for all $x \in X$.*

Proof. First assume that there is a non-trivial character χ with

$$\chi(T^n x) = \chi(x)$$

for some $n > 0$, chosen to be minimal with this property. Then the function

$$f(x) = \chi(x) + \chi(Tx) + \cdots + \chi(T^{n-1}x)$$

is invariant under T, and is non-constant since it is a sum of non-trivial distinct characters. It follows that T is not ergodic.

Conversely, assume that no non-trivial character is invariant under a non-zero power of T, and let $f \in L^2_{m_X}(X)$ be a function invariant under T. Then f has a Fourier expansion in $L^2_{m_X}$,

$$f = \sum_{\chi \in \widehat{X}} c_\chi \chi,$$

with $\sum_\chi |c_\chi|^2 = \|f\|_2^2 < \infty$. Since f is invariant, $c_\chi = c_{\chi \circ T} = c_{\chi \circ T^2} = \cdots$, so either $c_\chi = 0$ or there are only finitely many distinct characters among the $\chi \circ T^i$ (for otherwise $\sum_\chi |c_\chi|^2$ would be infinite). It follows that there are integers $p > q$ with $\chi \circ T^p = \chi \circ T^q$, which means that χ is invariant under T^{p-q} (the map $\chi \mapsto \chi \circ T$ from $\widehat{X}$ to $\widehat{X}$ is injective since T is surjective), so χ is trivial by hypothesis. It follows that the Fourier expansion of f is a constant, so T is ergodic. □

In particular, Theorem 2.19 may be applied to characterize ergodicity for endomorphisms of the torus.

Corollary 2.20. *Let $A \in \mathrm{Mat}_{dd}(\mathbb{Z})$ be an integer matrix with $\det(A) \neq 0$. Then A induces a surjective endomorphism T_A of $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ which preserves the Lebesgue measure $m_{\mathbb{T}^d}$. The transformation T_A is ergodic if and only if no eigenvalue of A is a root of unity.*

While harmonic analysis sometimes provides a short and readily understood proof of ergodic or mixing properties, these methods are in general less amenable to generalization than are the more geometric arguments.